Solutions to Axler's Linear Algebra

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Contents

3B Null Spaces and Ranges

Exercise 3B1

Give an example of a linear map T with $\dim \operatorname{null} T = 3$ and $\dim \operatorname{range} T = 2$.

Solution. Let $T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^2)$ be defined by

$$T(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2).$$

Then $\operatorname{null} T = \{(0, 0, x_3, x_4, x_5) \mid x_3, x_4, x_5 \in \mathbb{R}\}$, which has dimension 3, and $\operatorname{range} T = \mathbb{R}^2$, which has dimension 2. ■

Exercise 3B2

Suppose $S, T \in \mathcal{L}(V)$ are such that $\operatorname{range} S \subseteq \operatorname{null} T$. Prove that $(ST)^2 = 0$.

Solution. Let $v \in V$. Then

$$(ST)^2 v = ST(STv) = S(T(STv)).$$

Since $STv \in \operatorname{range} S$ and $\operatorname{range} S \subseteq \operatorname{null} T$, we have $T(STv) = 0$. Therefore,

$$(ST)^2 v = S(0) = 0.$$

Since v was arbitrary, it follows that $(ST)^2 = 0$. ■

Exercise 3B3

Suppose $v_1, \ldots, v_m$ is a list of vectors in V. Define $T \in \mathcal{L}(\mathbb{F}^m, V)$ by

$$T(z_1, \ldots, z_m) = z_1 v_1 + \cdots + z_m v_m.$$

(a) What property of T corresponds to $v_1, \ldots, v_m$ spanning V?

(b) What property of T corresponds to the list $v_1, \ldots, v_m$ being linearly independent?

Solution.

(a) The list $v_1, \ldots, v_m$ spans V if and only if $\operatorname{range} T = V$, i.e., T is surjective.

(b) The list $v_1, \ldots, v_m$ is linearly independent if and only if $\operatorname{null} T = \{0\}$, i.e., T is injective. ■

Exercise 3B4

Show that $\{T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4) \mid \dim \operatorname{null} T > 2\}$ is not a subspace of $\mathcal{L}(\mathbb{R}^5, \mathbb{R}^4)$.

Solution. Let $T_1, T_2 \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4)$ be defined by

$$T_1(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2, 0, 0)$$

and

$$T_2(x_1, x_2, x_3, x_4, x_5) = (0, 0, x_3, x_4).$$

Then $\operatorname{null} T_1 = \{(0, 0, x_3, x_4, x_5) \mid x_3, x_4, x_5 \in \mathbb{R}\}$ and $\operatorname{null} T_2 = \{(x_1, x_2, 0, 0, x_5) \mid x_1, x_2, x_5 \in \mathbb{R}\}$ both have dimension 3, so T_1 and T_2 are in the set $\{T \in \mathcal{L}(\mathbb{R}^5, \mathbb{R}^4) \mid \dim \operatorname{null} T > 2\}$. However,

$$(T_1 + T_2)(x_1, x_2, x_3, x_4, x_5) = (x_1, x_2, x_3, x_4),$$

which has null space $\operatorname{null}(T_1 + T_2) = \{(0, 0, 0, 0, x_5) \mid x_5 \in \mathbb{R}\}$ of dimension 1. Therefore,

$$\dim \operatorname{null}(T_1 + T_2) = 1 < 2,$$

so $T_1 + T_2$ is not in the set. Hence, the set is not closed under addition and is not a subspace of $\mathcal{L}(\mathbb{R}^5, \mathbb{R}^4)$. ■

Exercise 3B5

Give an example of $T \in \mathcal{L}(\mathbb{R}^4)$ such that $\operatorname{range} T = \operatorname{null} T$.

Solution. Let v_1, v_2, v_3, v_4 be the standard basis of $\mathbb{R}^4$. Define $T \in \mathcal{L}(\mathbb{R}^4)$ by

$$Tv_1 = v_3, \quad Tv_2 = v_4, \quad Tv_3 = 0, \quad Tv_4 = 0.$$

Then for any $x = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4$,

$$Tx = x_1 Tv_1 + x_2 Tv_2 + x_3 Tv_3 + x_4 Tv_4 = x_1 v_3 + x_2 v_4.$$

Then $\operatorname{range} T = \operatorname{span} v_3, v_4$. If $Tx = 0$, then $x_1 v_3 + x_2 v_4 = 0$, which implies $x_1 = x_2 = 0$. Therefore, $\operatorname{null} T = \{(0, 0, x_3, x_4) \mid x_3, x_4 \in \mathbb{R}\} = \operatorname{span} v_3, v_4$. Hence, $\operatorname{range} T = \operatorname{null} T$. ■

Exercise 3B6

Prove that there does not exist $T \in \mathcal{L}(\mathbb{R}^5)$ such that $\operatorname{range} T = \operatorname{null} T$.

Solution. Suppose for contradiction that there exists $T \in \mathcal{L}(\mathbb{R}^5)$ such that $\operatorname{range} T = \operatorname{null} T$. Then by the rank-nullity theorem,

$$\dim \mathbb{R}^5 = \dim \operatorname{null} T + \dim \operatorname{range} T = 2 \dim \operatorname{range} T.$$

Therefore, $5 = 2 \dim \operatorname{range} T$, which is impossible since 5 is odd and $2 \dim \operatorname{range} T$ is even. Hence, no such T exists. ■

Exercise 3B7

Suppose V and W are finite-dimensional with $2 \le \dim V \le \dim W$. Show that $\{T \in \mathcal{L}(V, W) \mid T \text{ is not injective}\}$ is not a subspace of $\mathcal{L}(V, W)$.

Solution. Let $\dim V = n$ and $\dim W = m$ with $2 \le n \le m$. Let $v_1, v_2, \ldots, v_n$ be a basis of V, and let $w_1, w_2, \ldots, w_m$ be a basis of W. Define $T_1, T_2 \in \mathcal{L}(V, W)$ by

$$T_1 v_1 = w_1, \quad T_1 v_k = 0 \text{ for } k \ge 2,$$

and

$$T_2 v_1 = 0, \quad T_2 v_k = w_k \text{ for } k \ge 2.$$

Then $v_2 \in \operatorname{null} T_1$ and $v_1 \in \operatorname{null} T_2$, so both T_1 and T_2 are not injective. However,

$$(T_1 + T_2)v_1 = T_1 v_1 + T_2 v_1 = w_1 + 0 = w_1 \neq 0,$$

and for $k \ge 2$,

$$(T_1 + T_2)v_k = T_1 v_k + T_2 v_k = 0 + w_k = w_k \neq 0.$$

Suppose $v \in V$ such that $(T_1 + T_2)v = 0$. Then v can be expressed as a linear combination of the basis vectors:

$$v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n.$$

Then

$$(T_1 + T_2)v = \alpha_1 w_1 + \alpha_2 w_2 + \cdots + \alpha_n w_n = 0.$$

Since $w_1, w_2, \ldots, w_m$ are linearly independent, it follows that $\alpha_1 = \alpha_2 = \cdots = \alpha_n = 0$, hence $v = 0$. Therefore, $T_1 + T_2$ is injective. Since T_1 and T_2 are not injective, but their sum is injective, the set $\{T \in \mathcal{L}(V, W) \mid T \text{ is not injective}\}$ is not closed under addition and thus is not a subspace of $\mathcal{L}(V, W)$. ■

Exercise 3B8

Suppose V and W are finite-dimensional with $\dim V \ge \dim W \ge 2$. Show that $\{T \in \mathcal{L}(V, W) \mid T \text{ is not surjective}\}$ is not a subspace of $\mathcal{L}(V, W)$.

Solution. We proceed similarly to the previous exercise. Let $\dim V = n$ and $\dim W = m$ with $n \ge m \ge 2$. Let $v_1, v_2, \ldots, v_n$ be a basis of V, and let $w_1, w_2, \ldots, w_m$ be a basis of W. Define $T_1, T_2 \in \mathcal{L}(V, W)$ by

$$T_1 v_1 = w_1, \quad T_1 v_k = 0 \text{ for } k \ge 2,$$

and

$$T_2 v_1 = 0, \quad T_2 v_k = w_k \text{ for } k = 2, \ldots, m, \quad T_2 v_k = 0 \text{ for } k > m.$$

Then $\operatorname{range} T_1 = \operatorname{span} w_1$ and $\operatorname{range} T_2 = \operatorname{span} w_2, \ldots, w_m$, both of which are proper subspaces of W, so T_1 and T_2 are not surjective. However,

$$(T_1 + T_2)v_1 = T_1 v_1 + T_2 v_1 = w_1 + 0 = w_1,$$

and for $k = 2, \ldots, m$,

$$(T_1 + T_2)v_k = T_1 v_k + T_2 v_k = 0 + w_k = w_k.$$

Therefore, $\operatorname{range}(T_1 + T_2) = \operatorname{span} w_1, w_2, \ldots, w_m = W$, so $T_1 + T_2$ is surjective. Since T_1 and T_2 are not surjective, but their sum is surjective, the set $\{T \in \mathcal{L}(V, W) \mid T \text{ is not surjective}\}$ is not closed under addition and thus is not a subspace of $\mathcal{L}(V, W)$. ■

Exercise 3B9

Suppose $T \in \mathcal{L}(V, W)$ is injective and $v_1, \ldots, v_n$ is linearly independent in V. Prove that $Tv_1, \ldots, Tv_n$ is linearly independent in W.

Solution. Suppose $a_1, \ldots, a_n \in \mathbb{F}$ are such that

$$a_1 Tv_1 + \cdots + a_n Tv_n = 0.$$

By linearity of T, we have

$$T(a_1 v_1 + \cdots + a_n v_n) = 0.$$

Injectivity of T implies that $\operatorname{null} T = \{0\}$, so

$$a_1 v_1 + \cdots + a_n v_n = 0.$$

Since $v_1, \ldots, v_n$ is linearly independent, it follows that $a_1 = \cdots = a_n = 0$. Therefore, $Tv_1, \ldots, Tv_n$ is linearly independent in W. ∎

Exercise 3B10

Suppose $v_1, \ldots, v_n$ spans V and $T \in \mathcal{L}(V, W)$. Show that $Tv_1, \ldots, Tv_n$ spans $\operatorname{range} T$.

Solution. Let $w \in \operatorname{range} T$. Then there exists $v \in V$ such that $Tv = w$. Since $v_1, \ldots, v_n$ spans V, we can write

$$v = a_1 v_1 + \cdots + a_n v_n$$

for some scalars $a_1, \ldots, a_n \in \mathbb{F}$. By linearity of T, we have

$$w = Tv = T(a_1 v_1 + \cdots + a_n v_n) = a_1 Tv_1 + \cdots + a_n Tv_n.$$

Therefore, w is a linear combination of $Tv_1, \ldots, Tv_n$. Since w was arbitrary in $\operatorname{range} T$, it follows that $Tv_1, \ldots, Tv_n$ spans $\operatorname{range} T$. ∎

Exercise 3B11

Suppose that V is finite-dimensional and that $T \in \mathcal{L}(V, W)$. Prove that there exists a subspace U of V such that

$$U \cap \operatorname{null} T = \{0\} \quad \text{and} \quad \operatorname{range} T = \{Tu \mid u \in U\}.$$

Manual Remark. *Note that this exercise proves the existence of a subspace U of V such that $V = U \oplus \operatorname{null} T$.*

Solution. Since $\operatorname{null} T$ is a subspace of V, we can extend a basis of $\operatorname{null} T$ to a basis of V. Let $n_1, \ldots, n_k$ be a basis of $\operatorname{null} T$, and extend this to a basis of V by adding vectors $u_1, \ldots, u_m$. Hence, a basis for V is given by

$$n_1, \ldots, n_k, u_1, \ldots, u_m.$$

Define $U = \operatorname{span}(u_1, \dots, u_m)$. Suppose $v \in U \cap \operatorname{null} T$. Then v can be expressed as a linear combination of the basis vectors of U:

$$v = \alpha_1 u_1 + \cdots + \alpha_m u_m.$$

Since $v \in \operatorname{null} T$ as well, we can also express v as a linear combination of the basis vectors of $\operatorname{null} T$:

$$v = \beta_1 n_1 + \cdots + \beta_k n_k.$$

Equating the two expressions for v, we have

$$\alpha_1 u_1 + \cdots + \alpha_m u_m = \beta_1 n_1 + \cdots + \beta_k n_k.$$

Since $n_1, \dots, n_k, u_1, \dots, u_m$ is a basis for V, the only way this equality can hold is if all coefficients are zero, i.e., $\alpha_1 = \cdots = \alpha_m = 0$ and $\beta_1 = \cdots = \beta_k = 0$. Therefore, $v = 0$, and we conclude that $U \cap \operatorname{null} T = \{0\}$.

Now, let $w \in \operatorname{range} T$. Then there exists $v \in V$ such that $Tv = w$. We can express v as a linear combination of the basis vectors of V:

$$v = \gamma_1 n_1 + \cdots + \gamma_k n_k + \delta_1 u_1 + \cdots + \delta_m u_m.$$

Applying T to both sides, we have

$$Tv = T(\gamma_1 n_1 + \cdots + \gamma_k n_k + \delta_1 u_1 + \cdots + \delta_m u_m) = \gamma_1 T n_1 + \cdots + \gamma_k T n_k + \delta_1 T u_1 + \cdots + \delta_m T u_m.$$

Since $n_1, \dots, n_k \in \operatorname{null} T$, we have $Tn_i = 0$ for all $i = 1, \dots, k$. Therefore,

$$Tv = \delta_1 T u_1 + \cdots + \delta_m T u_m.$$

Let $u = \delta_1 u_1 + \cdots + \delta_m u_m \in U$. Then $Tu = w$. Since w was arbitrary in $\operatorname{range} T$, it follows that $\operatorname{range} T = \{Tu \mid u \in U\}$. Thus, we have found a subspace U of V such that $U \cap \operatorname{null} T = \{0\}$ and $\operatorname{range} T = \{Tu \mid u \in U\}$. ■

Exercise 3B12

Suppose T is a linear map from $\mathbb{F}^4$ to $\mathbb{F}^2$ such that

$$\operatorname{null} T = \{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 \mid x_1 = 5x_2 \text{ and } x_3 = 7x_4\}.$$

Prove that T is surjective.

Solution. We may write the null space of T as

$$\operatorname{null} T = \{(5x_2, x_2, 7x_4, x_4) \mid x_2, x_4 \in \mathbb{F}\}.$$

From this, we can see that $\operatorname{null} T$ is spanned by the vectors $(5, 1, 0, 0)$ and $(0, 0, 7, 1)$. Therefore, $\dim \operatorname{null} T = 2$. By the rank-nullity theorem, we have

$$\dim \mathbb{F}^4 = \dim \operatorname{null} T + \dim \operatorname{range} T \implies 4 = 2 + \dim \operatorname{range} T.$$

Solving for $\dim \operatorname{range} T$, we find that $\dim \operatorname{range} T = 2$. Since $\dim \mathbb{F}^2 = 2$, it follows that $\operatorname{range} T = \mathbb{F}^2$. Therefore, T is surjective. ■

Exercise 3B13

Suppose U is a three-dimensional subspace of $\mathbb{R}^8$ and that T is a linear map from $\mathbb{R}^8$ to $\mathbb{R}^5$ such that $\operatorname{null} T = U$. Prove that T is surjective.

Solution. By the rank-nullity theorem, we have

$$\dim \mathbb{R}^8 = \dim \operatorname{null} T + \dim \operatorname{range} T \implies 8 = 3 + \dim \operatorname{range} T.$$

Solving for $\dim \operatorname{range} T$, we find that $\dim \operatorname{range} T = 5$. Since $\dim \mathbb{R}^5 = 5$, it follows that $\operatorname{range} T = \mathbb{R}^5$. Therefore, T is surjective. ■

Exercise 3B14

Prove that there does not exist a linear map from $\mathbb{F}^5$ to $\mathbb{F}^2$ whose null space equals $\{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{F}^5 \mid x_1 = 3x_2 \text{ and } x_3 = x_4 = x_5\}$.

Solution. We can express the null space as

$$\operatorname{null} T = \{(3x_2, x_2, x_4, x_4, x_4) \mid x_2, x_4 \in \mathbb{F}\}.$$

From this, we can see that $\operatorname{null} T$ is spanned by the vectors $(3, 1, 0, 0, 0)$ and $(0, 0, 1, 1, 1)$. Therefore, $\dim \operatorname{null} T = 2$. By the rank-nullity theorem, we have

$$\dim \mathbb{F}^5 = \dim \operatorname{null} T + \dim \operatorname{range} T \implies 5 = 2 + \dim \operatorname{range} T.$$

Solving for $\dim \operatorname{range} T$, we find that $\dim \operatorname{range} T = 3$. However, $\dim \mathbb{F}^2 = 2$, which is less than 3. This is a contradiction, as the dimension of the range cannot exceed the dimension of the target space. Therefore, no such linear map exists. ■

Exercise 3B15

Suppose there exists a linear map on V whose null space and range are both finite-dimensional. Prove that V is finite-dimensional.
Manual Remark. *The spanning list in the solution is actually a basis of V.*

Solution. Let $T \in \mathcal{L}(V)$ be such that $\operatorname{null} T$ and $\operatorname{range} T$ are both finite-dimensional. Let $u_1, \ldots, u_m$ be a basis of $\operatorname{null} T$, and let $w_1, \ldots, w_n$ be a basis of $\operatorname{range} T$. Select $v_j \in V$ be such that $Tv_j = w_j$ for $j = 1, \ldots, n$. We claim that the list $u_1, \ldots, u_m, v_1, \ldots, v_n$ spans V.

Let $v \in V$. Then $Tv \in \operatorname{range} T$, so we may write

$$Tv = \alpha_1 w_1 + \cdots + \alpha_n w_n$$

for some scalars $\alpha_1, \ldots, \alpha_n \in \mathbb{F}$. Consider the vector

$$u = v - (\alpha_1 v_1 + \cdots + \alpha_n v_n).$$

Then

$$Tu = Tv - (\alpha_1 Tv_1 + \cdots + \alpha_n Tv_n) = Tv - (\alpha_1 w_1 + \cdots + \alpha_n w_n) = 0.$$

Therefore, $u \in \operatorname{null} T$, and we can write

$$u = \beta_1 u_1 + \cdots + \beta_m u_m$$

for some scalars $\beta_1, \ldots, \beta_m \in \mathbb{F}$. Thus,

$$v = u + (\alpha_1 v_1 + \cdots + \alpha_n v_n) = \beta_1 u_1 + \cdots + \beta_m u_m + \alpha_1 v_1 + \cdots + \alpha_n v_n,$$

which shows that the list spans V. Hence, V is finite-dimensional. ■

Exercise 3B16

Suppose V and W are both finite-dimensional. Prove that there exists an injective linear map from V to W if and only if $\dim V \leq \dim W$.

Solution. Suppose there exists an injective linear map $T \in \mathcal{L}(V, W)$. By the rank-nullity theorem, we have

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T.$$

Since T is injective, $\operatorname{null} T = \{0\}$, so $\dim \operatorname{null} T = 0$. Therefore,

$$\dim V = 0 + \dim \operatorname{range} T = \dim \operatorname{range} T.$$

Since $\operatorname{range} T$ is a subspace of W, we have $\dim \operatorname{range} T \leq \dim W$. Thus,

$$\dim V = \dim \operatorname{range} T \leq \dim W.$$

Conversely, suppose $\dim V \leq \dim W$. Let $n = \dim V$ and $m = \dim W$. Choose a basis $\{v_1, \ldots, v_n\}$ of V. Let $\{w_1, \ldots, w_m\}$ be a basis of W. Define a linear map $T \in \mathcal{L}(V, W)$ by

$$Tv_i = w_i \quad \text{for } i = 1, \ldots, n.$$

Since $n \leq m$, we can assign each basis vector of V to a distinct basis vector of W. To show that T is injective, let $v = \alpha_1 v_1 + \cdots + \alpha_n v_n \in V$ such that $Tv = 0$. Then

$$Tv = \alpha_1 Tv_1 + \cdots + \alpha_n Tv_n = \alpha_1 w_1 + \cdots + \alpha_n w_n = 0.$$

Since $w_1, \ldots, w_n$ are linearly independent, it follows that $\alpha_1 = \cdots = \alpha_n = 0$. Therefore, $v = 0$, and T is injective. Hence, there exists an injective linear map from V to W if and only if $\dim V \leq \dim W$. ■

Exercise 3B17

Suppose V and W are both finite-dimensional. Prove that there exists a surjective linear map from V onto W if and only if $\dim V \geq \dim W$.

Solution. Suppose there exists a surjective linear map $T \in \mathcal{L}(V, W)$. By the rank-nullity theorem, we have

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T.$$

Since T is surjective, range $T = W$, so $\dim \operatorname{range} T = \dim W$. Therefore,

$$\dim V = \dim \operatorname{null} T + \dim W.$$

Since $\dim \operatorname{null} T \geq 0$, it follows that

$$\dim V \geq \dim W.$$

Conversely, suppose $\dim V \geq \dim W$. Let $n = \dim V$ and $m = \dim W$. Choose a basis $\{v_1, \ldots, v_n\}$ of V. Let $\{w_1, \ldots, w_m\}$ be a basis of W. Define a linear map $T \in \mathcal{L}(V, W)$ by

$$Tv_i = w_i \quad \text{for } i = 1, \ldots, m, \quad Tv_i = 0 \quad \text{for } i = m+1, \ldots, n.$$

To show that T is surjective, let $w \in W$. Then w can be expressed as a linear combination of the basis vectors of W:

$$w = \beta_1 w_1 + \cdots + \beta_m w_m.$$

Consider the vector $v = \beta_1 v_1 + \cdots + \beta_m v_m \in V$. Then

$$Tv = \beta_1 Tv_1 + \cdots + \beta_m Tv_m = \beta_1 w_1 + \cdots + \beta_m w_m = w.$$

Since w was arbitrary in W, it follows that range $T = W$. Therefore, T is surjective. Hence, there exists a surjective linear map from V onto W if and only if $\dim V \geq \dim W$. ■

Exercise 3B18

Suppose V and W are finite-dimensional and that U is a subspace of V. Prove that there exists $T \in \mathcal{L}(V, W)$ such that null $T = U$ if and only if $\dim U \geq \dim V - \dim W$.

Solution. Suppose there exists $T \in \mathcal{L}(V, W)$ such that null $T = U$. By the rank-nullity theorem, we have

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T = \dim U + \dim \operatorname{range} T.$$

Since range T is a subspace of W, we have $\dim \operatorname{range} T \leq \dim W$. Therefore,

$$\dim V = \dim U + \dim \operatorname{range} T \leq \dim U + \dim W.$$

Rearranging this inequality gives

$$\dim U \geq \dim V - \dim W.$$

Conversely, suppose $\dim U \geq \dim V - \dim W$. Let $n = \dim V$, $m = \dim W$, and $k = \dim U$. Let $u_1, \ldots, u_k$ be a basis of U. Extend this basis to a basis of V by adding vectors $v_1, \ldots, v_{n-k}$. Define a linear map $T \in \mathcal{L}(V, W)$ by

$$Tu_i = 0 \quad \text{for } i = 1, \ldots, k, \quad Tv_j = w_j \quad \text{for } j = 1, \ldots, n-k,$$

where $w_1, \ldots, w_{n-k}$ are linearly independent vectors in W. Since $\dim U = k$ and $\dim V - \dim W = n - m$, we have $k \geq n - m$, which implies that $n - k \leq m$. Therefore, we can choose $w_1, \ldots, w_{n-k}$ to be linearly independent in W. To show that null $T = U$, let $v \in \operatorname{null} T$. Then $Tv = 0$. Express v as a linear combination of the basis vectors of V:

$$v = \alpha_1 u_1 + \cdots + \alpha_k u_k + \beta_1 v_1 + \cdots + \beta_{n-k} v_{n-k}.$$

Applying T to both sides, we have

$$Tv = \alpha_1 Tu_1 + \cdots + \alpha_k Tu_k + \beta_1 Tv_1 + \cdots + \beta_{n-k} Tv_{n-k} = 0 + \beta_1 w_1 + \cdots + \beta_{n-k} w_{n-k} = 0.$$

Since $w_1, \ldots, w_{n-k}$ are linearly independent, it follows that $\beta_1 = \cdots = \beta_{n-k} = 0$. Therefore,

$$v = \alpha_1 u_1 + \cdots + \alpha_k u_k \in U.$$

Hence, $\operatorname{null} T \subseteq U$. Conversely, if $v \in U$, then $Tv = 0$ by the definition of T. Therefore, $U \subseteq \operatorname{null} T$. Combining these two inclusions, we have $\operatorname{null} T = U$. Thus, there exists $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = U$ if and only if $\dim U \geq \dim V - \dim W$. ■

Exercise 3B19

Suppose W is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that T is injective if and only if there exists $S \in \mathcal{L}(W, V)$ such that ST is the identity operator on V.

Solution. Suppose T is injective. Then $\operatorname{null} T = \{0\}$. Also, $\operatorname{range} T$ is finite dimensional since it is a subspace of W. Since T is injective, it is an isomorphism from V onto $\operatorname{range} T$. Hence, V is finite dimensional. Let $n = \dim V$ and $m = \dim W$. Since T is injective, we have $\dim \operatorname{range} T = n$. Choose a basis $\{v_1, \ldots, v_n\}$ of V. Then $\{Tv_1, \ldots, Tv_n\}$ is linearly independent in W. Extend this set to a basis of W by adding vectors $w_1, \ldots, w_{m-n}$. Define a linear map $S \in \mathcal{L}(W, V)$ by

$$S(Tv_i) = v_i \quad \text{for } i = 1, \ldots, n, \quad S(w_j) = 0 \quad \text{for } j = 1, \ldots, m - n.$$

Then for any $v \in V$, we can express v as a linear combination of the basis vectors:

$$v = \alpha_1 v_1 + \cdots + \alpha_n v_n.$$

Applying T to both sides, we have

$$Tv = \alpha_1 Tv_1 + \cdots + \alpha_n Tv_n.$$

Now, applying S to Tv, we get

$$STv = S(\alpha_1 Tv_1 + \cdots + \alpha_n Tv_n) = \alpha_1 S(Tv_1) + \cdots + \alpha_n S(Tv_n) = \alpha_1 v_1 + \cdots + \alpha_n v_n = v.$$

Therefore, ST is the identity operator on V.

Conversely, suppose there exists $S \in \mathcal{L}(W, V)$ such that ST is the identity operator on V. Let $v \in \operatorname{null} T$. Then $Tv = 0$, and applying S gives

$$STv = S(0) = 0.$$

Since ST is the identity operator on V, we have $v = STv = 0$. Therefore, $\operatorname{null} T = \{0\}$, which means T is injective. ■

Exercise 3B20

Suppose W is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that T is surjective if and only if there exists $S \in \mathcal{L}(W, V)$ such that TS is the identity operator on W.

Solution. Suppose T is surjective. Then $\operatorname{range} T = W$. Let $m = \dim W$ and choose a basis $\{w_1, \ldots, w_m\}$ of W. Since T is surjective, for each w_i, there exists $v_i \in V$ such that $Tv_i = w_i$. Define a linear map $S \in \mathcal{L}(W, V)$ by

$$S(w_i) = v_i \quad \text{for } i = 1, \ldots, m.$$

Then for any $w \in W$, we can express w as a linear combination of the basis vectors:

$$w = \alpha_1 w_1 + \cdots + \alpha_m w_m.$$

Applying S to both sides, we have

$$Sw = S(\alpha_1 w_1 + \cdots + \alpha_m w_m) = \alpha_1 S(w_1) + \cdots + \alpha_m S(w_m) = \alpha_1 v_1 + \cdots + \alpha_m v_m.$$

Now, applying T to Sw, we get

$$TSw = T(\alpha_1 v_1 + \cdots + \alpha_m v_m) = \alpha_1 Tv_1 + \cdots + \alpha_m Tv_m = \alpha_1 w_1 + \cdots + \alpha_m w_m = w.$$

Therefore, TS is the identity operator on W.

Conversely, suppose there exists $S \in \mathcal{L}(W, V)$ such that TS is the identity operator on W. Let $w \in W$. Then

$$TSw = w.$$

This means that for every $w \in W$, there exists a vector $v = Sw \in V$ such that $Tv = T(Sw) = w$. Therefore, $\operatorname{range} T = W$, which means that T is surjective. ■

Exercise 3B21

Suppose V is finite-dimensional, $T \in \mathcal{L}(V, W)$, and U is a subspace of W. Prove that $\{v \in V \mid Tv \in U\}$ is a subspace of V and

$$\dim\{v \in V \mid Tv \in U\} = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T).$$

Solution. Let $X = \{v \in V \mid Tv \in U\}$. We first show that X is a subspace of V. Let $v_1, v_2 \in X$ and $\alpha, \beta \in \mathbb{F}$. Then

$$T(\alpha v_1 + \beta v_2) = \alpha Tv_1 + \beta Tv_2.$$

Since $Tv_1, Tv_2 \in U$ and U is a subspace of W, it follows that $\alpha Tv_1 + \beta Tv_2 \in U$. Therefore, $\alpha v_1 + \beta v_2 \in X$, and thus X is a subspace of V.

Now, we will compute the dimension of X. Let $Y = U \cap \operatorname{range} T$. Let $\dim T = k$ and $\dim Y = m$. Let $n_1, \ldots, n_k$ be a basis of $\operatorname{null} T$, and let $y_1, \ldots, y_m$ be a basis of Y. Since $y_i \in \operatorname{range} T$, there exist vectors $v_i \in V$ such that $Tv_i = y_i$ for $i = 1, \ldots, m$. We claim that the set $\{n_1, \ldots, n_k, v_1, \ldots, v_m\}$ is a basis of X.

First, we show that this set spans X. Let $v \in X$. Then $Tv \in U$, and since $Tv \in \operatorname{range} T$, we have $Tv \in Y$. Therefore, there exist scalars $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$ such that

$$Tv = \alpha_1 y_1 + \cdots + \alpha_m y_m = \alpha_1 Tv_1 + \cdots + \alpha_m Tv_m.$$

Let $v' = v - (\alpha_1 v_1 + \cdots + \alpha_m v_m)$. Then

$$Tv' = Tv - T(\alpha_1 v_1 + \cdots + \alpha_m v_m) = Tv - (\alpha_1 Tv_1 + \cdots + \alpha_m Tv_m) = 0.$$

Therefore, $v' \in \operatorname{null} T$, and we can express v' as a linear combination of the basis vectors of $\operatorname{null} T$:

$$v' = \beta_1 n_1 + \cdots + \beta_k n_k$$

for some scalars $\beta_1, \ldots, \beta_k \in \mathbb{F}$. Thus,

$$v = v' + (\alpha_1 v_1 + \cdots + \alpha_m v_m) = \beta_1 n_1 + \cdots + \beta_k n_k + \alpha_1 v_1 + \cdots + \alpha_m v_m.$$

This shows that v is a linear combination of the vectors in $\{n_1, \ldots, n_k, v_1, \ldots, v_m\}$, and hence this set spans X.

Next, we show that this set is linearly independent. Suppose

$$\gamma_1 n_1 + \cdots + \gamma_k n_k + \delta_1 v_1 + \cdots + \delta_m v_m = 0$$

for some scalars $\gamma_1, \ldots, \gamma_k, \delta_1, \ldots, \delta_m \in \mathbb{F}$. Observe that $Tn_i = 0$ for all $i = 1, \ldots, k$, and $Tv_j = y_j$ for all $j = 1, \ldots, m$. Therefore,

$$0 = T(\gamma_1 n_1 + \cdots + \gamma_k n_k + \delta_1 v_1 + \cdots + \delta_m v_m) = \delta_1 y_1 + \cdots + \delta_m y_m.$$

Since $y_1, \ldots, y_m$ are linearly independent, it follows that $\delta_1 = \cdots = \delta_m = 0$. Therefore,

$$\gamma_1 n_1 + \cdots + \gamma_k n_k = 0.$$

Since $n_1, \ldots, n_k$ are linearly independent, then $\gamma_1 = \cdots = \gamma_k = 0$. Thus, the set $\{n_1, \ldots, n_k, v_1, \ldots, v_m\}$ is linearly independent and a basis of X. Therefore,

$$\dim X = k + m = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T).$$

■

Exercise 3B22

Suppose U and V are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(U, V)$. Prove that

$$\dim \operatorname{null} ST \leq \dim \operatorname{null} S + \dim \operatorname{null} T.$$

Solution. Let $u \in U$ such that $u \in \operatorname{null} ST$. Then $STu = 0$, which implies that $Tu \in \operatorname{null} S$. Therefore, we can express $\operatorname{null} ST$ as

$$\operatorname{null} ST = \{u \in U \mid Tu \in \operatorname{null} S\}.$$

Observe that this looks similar to the set in Exercise 3b21, where T is a linear map from U to V and $\operatorname{null} S$ is a subspace of V. By applying the result from Exercise 3b21, we have

$$\dim \operatorname{null} ST = \dim \operatorname{null} T + \dim(\operatorname{null} S \cap \operatorname{range} T).$$

Since $\operatorname{null} S \cap \operatorname{range} T$ is a subspace of $\operatorname{null} S$, we have

$$\dim(\operatorname{null} S \cap \operatorname{range} T) \leq \dim \operatorname{null} S.$$

Therefore,

$$\dim \operatorname{null} ST = \dim \operatorname{null} T + \dim(\operatorname{null} S \cap \operatorname{range} T) \leq \dim \operatorname{null} T + \dim \operatorname{null} S. \quad ■$$

Exercise 3B23

Suppose U and V are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(U, V)$. Prove that
$$\dim \operatorname{range} ST \leq \min(\dim \operatorname{range} S, \dim \operatorname{range} T).$$

Solution. Let $u \in U$ such that $STu \in \operatorname{range} ST$. Then $STu = S(Tu)$, which implies that $Tu \in \operatorname{range} T$. Therefore, we can express range ST as

$$\operatorname{range} ST = \{S(Tu) \mid u \in U\}.$$

Since $Tu \in \operatorname{range} T$, we can rewrite this as

$$\operatorname{range} ST = \{Sv \mid v \in \operatorname{range} T\}.$$

This shows that range ST is a subset of range S, and hence

$$\dim \operatorname{range} ST \leq \dim \operatorname{range} S.$$

Now let $v_1, \ldots, v_m$ be a basis of range T. Then for any $u \in U$, we can express Tu as a linear combination of the basis vectors:
$$Tu = \alpha_1 v_1 + \cdots + \alpha_m v_m$$
for some scalars $\alpha_1, \ldots, \alpha_m \in \mathbb{F}$. Applying S to both sides, we have

$$STu = S(Tu) = S(\alpha_1 v_1 + \cdots + \alpha_m v_m) = \alpha_1 S v_1 + \cdots + \alpha_m S v_m.$$

This shows that range ST is spanned by the vectors $Sv_1, \ldots, Sv_m$. Therefore,

$$\dim \operatorname{range} ST \leq m = \dim \operatorname{range} T.$$

Combining the two inequalities, we have

$$\dim \operatorname{range} ST \leq \min(\dim \operatorname{range} S, \dim \operatorname{range} T). \qquad \blacksquare$$

Exercise 3B24

(a) Suppose $\dim V = 5$ and $S, T \in \mathcal{L}(V)$ are such that $ST = 0$. Prove that $\dim \operatorname{range} TS \leq 2$.

(b) Give an example of $S, T \in \mathcal{L}(\mathbb{F}^5)$ with $ST = 0$ and $\dim \operatorname{range} TS = 2$.

Solution.

(a) Since $ST = 0$, then $S(Tv) = 0$ for all $v \in V$. This implies that range $T \subseteq$ null S. By the rank-nullity theorem, we have

$$5 = \dim \operatorname{null} S + \dim \operatorname{range} S \geq \dim \operatorname{range} T + \dim \operatorname{range} S.$$

Now, by Exercise 3B23, we have

$$\dim \operatorname{range} TS \leq \min(\dim \operatorname{range} T, \dim \operatorname{range} S).$$

Suppose that $\dim \operatorname{range} TS > 2$. Then both $\dim \operatorname{range} T$ and $\dim \operatorname{range} S$ must be greater than 2, or equivalently, $\dim \operatorname{range} T \geq 3$ and $\dim \operatorname{range} S \geq 3$. This implies that

$$\dim \operatorname{range} T + \dim \operatorname{range} S \geq 3 + 3 = 6 > 5,$$

which is a contradiction. Therefore, $\dim \operatorname{range} TS \leq 2$.

(b) Let v_1, v_2, v_3, v_4, v_5 be a basis of $\mathbb{F}^5$. Define $S, T \in \mathcal{L}(\mathbb{F}^5)$ by

$$Sv_1 = v_1, \quad Sv_2 = v_2, \quad Sv_3 = Sv_4 = Sv_5 = 0,$$

and

$$Tv_1 = v_3, \quad Tv_2 = v_4, \quad Tv_3 = Tv_4 = Tv_5 = 0.$$

Then $ST = 0$ since $S(Tv_i) = S(0) = 0$ for all $i = 1, \dots, 5$. Now, we compute $\operatorname{range} TS$. We have

$$TSv_1 = T(Sv_1) = Tv_1 = v_3, \quad TSv_2 = T(Sv_2) = Tv_2 = v_4, \quad TSv_3 = TSv_4 = TSv_5 = 0.$$

Therefore, $\operatorname{range} TS = \operatorname{span}(v_3, v_4)$, which has dimension 2. Thus, we have constructed an example of $S, T \in \mathcal{L}(\mathbb{F}^5)$ with $ST = 0$ and $\dim \operatorname{range} TS = 2$. ■

Exercise 3B25

Suppose that W is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{null} S \subseteq \operatorname{null} T$ if and only if there exists $E \in \mathcal{L}(W)$ such that $T = ES$.

Manual Remark. *This exercise requires showing that some $R \in \mathcal{L}(\operatorname{range} S, W)$ is well-defined. An issue arises when T is not injective: for some $w \in \operatorname{range} S$, there may be distinct $v_1, v_2 \in V$ such that $Sv_1 = Sv_2 = w$. In this case, we must show that $Tv_1 = Tv_2$.*

Solution. Suppose $\operatorname{null} S \subseteq \operatorname{null} T$. Define a linear map $R : \operatorname{range} S \to W$ by $R(Sv) = Tv$ for all $v \in V$. We need to show that R is well-defined. Suppose $Sv_1 = Sv_2$ for some $v_1, v_2 \in V$. Then $S(v_1 - v_2) = 0$, which implies that $v_1 - v_2 \in \operatorname{null} S$. Since $\operatorname{null} S \subseteq \operatorname{null} T$, we have $v_1 - v_2 \in \operatorname{null} T$, which means that $T(v_1 - v_2) = 0$. Therefore, $Tv_1 = Tv_2$, and hence $R(Sv_1) = R(Sv_2)$. Thus, R is well-defined.

By Exercise 3A13, we can extend R to some $E \in \mathcal{L}(W)$. Then for any $v \in V$, we have

$$Tv = R(Sv) = E(Sv),$$

and hence $T = ES$.

Conversely, suppose there exists $E \in \mathcal{L}(W)$ such that $T = ES$. Let $v \in \operatorname{null} S$. Then $Sv = 0$, and applying E gives

$$Tv = E(Sv) = E(0) = 0.$$

Therefore, $v \in \operatorname{null} T$, and hence $\operatorname{null} S \subseteq \operatorname{null} T$. ■

Exercise 3B26

Suppose that V is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{range} S \subseteq \operatorname{range} T$ if and only if there exists $E \in \mathcal{L}(V)$ such that $S = TE$.

Manual Remark. *From the previous exercise, the reader may be tempted to show that a map is well-defined again. However, this is not necessary here, since we are defining a map on the entire space V, not just on $\operatorname{range} S$.*

Solution. Suppose range $S \subseteq$ range T. Let $v_1, \ldots, v_n$ be a basis of V. For each $i = 1, \ldots, n$, since $Sv_i \in$ range $S \subseteq$ range T, there exists $u_i \in V$ such that $Tu_i = Sv_i$. Define a linear map $E \in \mathcal{L}(V)$ by

$$Ev_i = u_i \quad \text{for } i = 1, \ldots, n.$$

Then for any $v \in V$, we can express v as a linear combination of the basis vectors:

$$v = \alpha_1 v_1 + \cdots + \alpha_n v_n$$

for some scalars $\alpha_1, \ldots, \alpha_n \in \mathbb{F}$. Applying S and TE to both sides, we have

$$Sv = \alpha_1 Sv_1 + \cdots + \alpha_n Sv_n = \alpha_1 Tu_1 + \cdots + \alpha_n Tu_n = T(\alpha_1 u_1 + \cdots + \alpha_n u_n) = TEv.$$

Therefore, $S = TE$.

Conversely, suppose there exists $E \in \mathcal{L}(V)$ such that $S = TE$. Let $w \in$ range S. Then there exists $v \in V$ such that $Sv = w$. Applying the definition of S, we have

$$w = Sv = TE(v) = T(E(v)).$$

Therefore, $w \in$ range T, and hence range $S \subseteq$ range T. ∎

Exercise 3B27

Suppose $P \in \mathcal{L}(V)$ and $P^2 = P$. Prove that $V = \operatorname{null} P \oplus \operatorname{range} P$.

Solution. Let $v \in V$. We can express v as

$$v = (v - Pv) + Pv.$$

Observe that $Pv \in$ range P by definition. Now, we need to show that $v - Pv \in$ null P. Applying P to $v - Pv$, we have

$$P(v - Pv) = Pv - P^2 v = Pv - Pv = 0.$$

Therefore, $v - Pv \in$ null P. This shows that every vector $v \in V$ can be expressed as a sum of a vector in null P and a vector in range P, which implies that

$$V = \operatorname{null} P + \operatorname{range} P.$$

Next, we need to show that the sum is direct. Suppose $u \in \operatorname{null} P \cap \operatorname{range} P$. Then there exists some $w \in V$ such that $u = Pw$. Since $u \in$ null P, we have

$$Pu = P(Pw) = P^2 w = Pw = u.$$

But since $u \in$ null P, we also have $Pu = 0$. Therefore, we have

$$u = Pu = 0.$$

This shows that the intersection of null P and range P is trivial, i.e., $\operatorname{null} P \cap \operatorname{range} P = \{0\}$. Hence, the sum is direct, and we conclude that

$$V = \operatorname{null} P \oplus \operatorname{range} P.$$ ∎

Exercise 3B28

Suppose $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}))$ is such that $\deg Dp = (\deg p) - 1$ for every nonconstant polynomial $p \in \mathcal{P}(\mathbb{R})$. Prove that D is surjective.

Manual Remark. *Note that $\mathcal{P}(\mathbb{R})$ is infinite-dimensional, so we cannot use the rank-nullity theorem directly. Instead, one should consider finite-dimensional subspaces of $\mathcal{P}(\mathbb{R})$.*

Solution. Let $p \in \mathcal{P}(\mathbb{R})$ be an arbitrary polynomial. We want to show that there exists a polynomial $q \in \mathcal{P}(\mathbb{R})$ such that $Dq = p$. Consider the subspace $\mathcal{P}_{n+1}(\mathbb{R})$ of polynomials of degree at most $n+1$, where $n = \deg p$. Since $\deg Dq = (\deg q) - 1$, we have $D : \mathcal{P}_{n+1}(\mathbb{R}) \to \mathcal{P}_n(\mathbb{R})$. The dimension of $\mathcal{P}_{n+1}(\mathbb{R})$ is $n+2$, and the dimension of $\mathcal{P}_n(\mathbb{R})$ is $n+1$. By the rank-nullity theorem, we have

$$\dim \operatorname{null} D + \dim \operatorname{range} D = \dim \mathcal{P}_{n+1}(\mathbb{R}) = n + 2.$$

Since $\deg Dp = (\deg p) - 1$ for every nonconstant polynomial p, the kernel of D consists of constant polynomials, which has dimension 1. Therefore, $\dim \operatorname{null} D = 1$, and we have

$$\dim \operatorname{range} D = n + 2 - 1 = n + 1 = \dim \mathcal{P}_n(\mathbb{R}).$$

Since $\dim \operatorname{range} D = \dim \mathcal{P}_n(\mathbb{R})$, it follows that $\operatorname{range} D = \mathcal{P}_n(\mathbb{R})$. Therefore, for any polynomial $p \in \mathcal{P}_n(\mathbb{R})$, there exists a polynomial $q \in \mathcal{P}_{n+1}(\mathbb{R})$ such that $Dq = p$. Since p was arbitrary, we conclude that D is surjective. ■

Exercise 3B29

Suppose $p \in \mathcal{P}(\mathbb{R})$. Prove that there exists a polynomial $q \in \mathcal{P}(\mathbb{R})$ such that $5q'' + 3q' = p$.

Solution. Let $p \in \mathcal{P}(\mathbb{R})$ be an arbitrary polynomial. We want to show that there exists a polynomial $q \in \mathcal{P}(\mathbb{R})$ such that $5q'' + 3q' = p$. Consider the subspace $\mathcal{P}_{n+2}(\mathbb{R})$ of polynomials of degree at most $n+2$, where $n = \deg p$. Define the linear map $D : \mathcal{P}_{n+2}(\mathbb{R}) \to \mathcal{P}_{n+1}(\mathbb{R})$ by $Dq = 5q'' + 3q'$. The dimension of $\mathcal{P}_{n+2}(\mathbb{R})$ is $n+3$, and the dimension of $\mathcal{P}_{n+1}(\mathbb{R})$ is $n+2$. By the rank-nullity theorem, we have

$$\dim \operatorname{null} D + \dim \operatorname{range} D = \dim \mathcal{P}_{n+2}(\mathbb{R}) = n + 3.$$

The kernel of D consists of polynomials q such that $5q'' + 3q' = 0$, or $5q'' = -3q'$. Note that if $\deg q = m$, then $\deg q' = m - 1$ and $\deg q'' = m - 2$. Therefore, the degree of $5q'' + 3q'$ is at most $m - 1$. For this to be zero, we must have $q' = 0$, which implies that q is a constant polynomial. Therefore, $\dim \operatorname{null} D = 1$, and we have

$$\dim \operatorname{range} D = n + 3 - 1 = n + 2 = \dim \mathcal{P}_{n+1}(\mathbb{R}).$$

Since $\dim \operatorname{range} D = \dim \mathcal{P}_{n+1}(\mathbb{R})$, it follows that $\operatorname{range} D = \mathcal{P}_{n+1}(\mathbb{R})$. Therefore, for any polynomial $p \in \mathcal{P}_{n+1}(\mathbb{R})$, there exists a polynomial $q \in \mathcal{P}_{n+2}(\mathbb{R})$ such that $Dq = p$. Since p was arbitrary, we conclude that D is surjective. ■

Exercise 3B30

Suppose $\varphi \in \mathcal{L}(V, \mathbb{F})$ and $\varphi \neq 0$. Suppose $u \in V$ is not in $\operatorname{null} \varphi$. Prove that

$$V = \operatorname{null} \varphi \oplus \{au \mid a \in \mathbb{F}\}.$$

Solution. Let $v \in V$. We want to show that v can be expressed as a sum of a vector in $\operatorname{null}\varphi$ and a scalar multiple of u. Since $\varphi(u) \neq 0$, we can define a scalar

$$a = \frac{\varphi(v)}{\varphi(u)}.$$

Then consider the vector

$$w = v - au.$$

We have

$$\varphi(w) = \varphi(v - au) = \varphi(v) - a\varphi(u) = \varphi(v) - \frac{\varphi(v)}{\varphi(u)}\varphi(u) = 0.$$

Therefore, $w \in \operatorname{null}\varphi$. Thus, we can express v as

$$v = w + au,$$

where $w \in \operatorname{null}\varphi$ and au is a scalar multiple of u. This shows that every vector in V can be expressed as a sum of a vector in $\operatorname{null}\varphi$ and a scalar multiple of u, which implies that

$$V = \operatorname{null}\varphi + \{au \mid a \in \mathbb{F}\}.$$

Next, we need to show that the sum is direct. Suppose $w \in \operatorname{null}\varphi \cap \{au \mid a \in \mathbb{F}\}$. Then there exists a scalar $a \in \mathbb{F}$ such that $w = au$. Since $w \in \operatorname{null}\varphi$, we have

$$0 = \varphi(w) = \varphi(au) = a\varphi(u).$$

Since $\varphi(u) \neq 0$, it follows that $a = 0$. Therefore, $w = 0$, which shows that the intersection of $\operatorname{null}\varphi$ and $\{au \mid a \in \mathbb{F}\}$ is trivial. Hence, the sum is direct, and we conclude that

$$V = \operatorname{null}\varphi \oplus \{au \mid a \in \mathbb{F}\}.$$ ■

Exercise 3B31

Suppose V is finite-dimensional, X is a subspace of V, and Y is a finite-dimensional subspace of W. Prove that there exists $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = X$ and $\operatorname{range} T = Y$ if and only if $\dim X + \dim Y = \dim V$.

Solution. Suppose there exists $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = X$ and $\operatorname{range} T = Y$. By the rank-nullity theorem, we have

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T = \dim X + \dim Y.$$

Conversely, suppose $\dim X + \dim Y = \dim V$. Let $n = \dim V$, $k = \dim X$, and $m = \dim Y$. Then $n = k + m$. Choose a basis $\{x_1, \dots, x_k\}$ of X and extend it to a basis $\{x_1, \dots, x_k, v_1, \dots, v_m\}$ of V. Let $\{y_1, \dots, y_m\}$ be a basis of Y. Define a linear map $T \in \mathcal{L}(V, W)$ by

$$Tx_i = 0 \quad \text{for } i = 1, \dots, k,$$

and

$$Tv_j = y_j \quad \text{for } j = 1, \dots, m.$$

Then for any $v \in V$, we can express v as a linear combination of the basis vectors:

$$v = \alpha_1 x_1 + \cdots + \alpha_k x_k + \beta_1 v_1 + \cdots + \beta_m v_m$$

for some scalars $\alpha_1, \ldots, \alpha_k, \beta_1, \ldots, \beta_m \in \mathbb{F}$. Applying T to both sides gives

$$Tv = \alpha_1 Tx_1 + \cdots + \alpha_k Tx_k + \beta_1 Tv_1 + \cdots + \beta_m Tv_m = 0 + \beta_1 y_1 + \cdots + \beta_m y_m.$$

Therefore, $Tv \in Y$. Now suppose $y = \gamma_1 y_1 + \cdots + \gamma_m y_m \in Y$. Then

$$T(\gamma_1 v_1 + \cdots + \gamma_m v_m) = \gamma_1 y_1 + \cdots + \gamma_m y_m = y.$$

Hence, $\operatorname{range} T = Y$. Moreover, if $Tv = 0$, then $\beta_1 y_1 + \cdots + \beta_m y_m = 0$. Since $y_1, \ldots, y_m$ are linearly independent, it follows that $\beta_1 = \cdots = \beta_m = 0$. Thus, $v \in X$. Additionally, if $v \in X$, then $v = \alpha_1 x_1 + \cdots + \alpha_k x_k$ for some scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$, and hence $Tv = 0$. Therefore, $\operatorname{null} T = X$. This shows that there exists a linear map $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = X$ and $\operatorname{range} T = Y$. ■

Exercise 3B32

Suppose V is finite-dimensional with $\dim V > 1$. Show that if $\varphi : \mathcal{L}(V) \to \mathbb{F}$ is a linear map such that $\varphi(ST) = \varphi(S)\varphi(T)$ for all $S, T \in \mathcal{L}(V)$, then $\varphi = 0$.
Axler Note. *The description of the two-sided ideals of $\mathcal{L}(V)$ given by Exercise 3A17 might be useful.*

Solution. Observe that $\operatorname{null} \varphi$ is a two-sided ideal of $\mathcal{L}(V)$ as follows: if $X \in \operatorname{null} \varphi$ and $Y \in \mathcal{L}(V)$, then

$$\varphi(XY) = \varphi(X)\varphi(Y) = 0 \cdot \varphi(Y) = 0, \quad \varphi(YX) = \varphi(Y)\varphi(X) = \varphi(Y) \cdot 0 = 0.$$

Since $\dim V > 1$, then Exercise 3A16 gives the existence of $S, T \in \mathcal{L}(V)$ such that $ST \neq TS$. Then we have

$$\varphi(ST - TS) = \varphi(S)\varphi(T) - \varphi(T)\varphi(S) = 0.$$

Therefore, $ST - TS \in \operatorname{null} \varphi$. Since $\operatorname{null} \varphi$ is a two-sided ideal of $\mathcal{L}(V)$, then by Exercise 3A17, we have $\operatorname{null} \varphi = \mathcal{L}(V)$. Hence, $\varphi = 0$. ■

Exercise 3B33

Suppose that V and W are real vector spaces and $T \in \mathcal{L}(V, W)$. Define $T_{\mathbb{C}} : V_{\mathbb{C}} \to W_{\mathbb{C}}$ by

$$T_{\mathbb{C}}(u + iv) = Tu + iTv$$

for all $u, v \in V$.

(a) Show that $T_{\mathbb{C}}$ is a (complex) linear map from $V_{\mathbb{C}}$ to $W_{\mathbb{C}}$.

(b) Show that $T_{\mathbb{C}}$ is injective if and only if T is injective.

(c) Show that $\operatorname{range} T_{\mathbb{C}} = W_{\mathbb{C}}$ if and only if $\operatorname{range} T = W$.

Axler Note. *See Exercise 8 in Section 1B for the definition of the complexification $V_{\mathbb{C}}$. The linear map $T_{\mathbb{C}}$ is called the **complexification** of the linear map T.*

Solution.

(a) Let $u + vi$ and $w + xi \in V_{\mathbb{C}}$. Then for any $a + bi \in \mathbb{C}$, we have

$$\begin{aligned} T_{\mathbb{C}}((a+bi)(u+vi)+(w+xi)) &= T_{\mathbb{C}}((au-bv+(av+bu)i)+(w+xi)) \\ &= T(au-bv+w)+iT(av+bu+x) \\ &= aTu-bTv+Tw+i(aTv+bTu+Tx) \\ &= (a+bi)(Tu+iTv)+(Tw+iTx) \\ &= (a+bi)T_{\mathbb{C}}(u+vi)+T_{\mathbb{C}}(w+xi). \end{aligned}$$

Therefore, $T_{\mathbb{C}}$ is a complex linear map from $V_{\mathbb{C}}$ to $W_{\mathbb{C}}$.

(b) Suppose $T_{\mathbb{C}}$ is injective. Let $v \in V$ such that $Tv = 0$. Then

$$T_{\mathbb{C}}(v+i0) = Tv + iT0 = 0 + i0 = 0.$$

Since $T_{\mathbb{C}}$ is injective, we have $v + i0 = 0$, which implies that $v = 0$. Therefore, T is injective.

Conversely, suppose T is injective. Let $u + iv \in V_{\mathbb{C}}$ such that $T_{\mathbb{C}}(u + iv) = 0$. Then

$$Tu + iTv = 0.$$

This implies that $Tu = 0$ and $Tv = 0$. Since T is injective, we have $u = 0$ and $v = 0$. Therefore, $u + iv = 0$, and hence $T_{\mathbb{C}}$ is injective.

(c) Suppose $\operatorname{range} T_{\mathbb{C}} = W_{\mathbb{C}}$. Let $w \in W$. Then $w + i0 \in W_{\mathbb{C}}$, and since $\operatorname{range} T_{\mathbb{C}} = W_{\mathbb{C}}$, there exists $u + iv \in V_{\mathbb{C}}$ such that

$$T_{\mathbb{C}}(u+iv) = Tu + iTv = w + i0.$$

This implies that $Tu = w$ and $Tv = 0$. Therefore, $w \in \operatorname{range} T$, and hence $\operatorname{range} T = W$.

Conversely, suppose $\operatorname{range} T = W$. Let $w_1 + iw_2 \in W_{\mathbb{C}}$. Since $\operatorname{range} T = W$, there exist $u, v \in V$ such that $Tu = w_1$ and $Tv = w_2$. Then

$$T_{\mathbb{C}}(u+iv) = Tu + iTv = w_1 + iw_2.$$

Therefore, $w_1 + iw_2 \in \operatorname{range} T_{\mathbb{C}}$, and hence $\operatorname{range} T_{\mathbb{C}} = W_{\mathbb{C}}$. ■